Fixed-margin prediction of RNA-protein association across cohorts

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Abstract

Background: Linked single-cell assays resolve cross-modal dependence because they measure molecular states in the same cell. Recipient cohorts may provide the marginal state frequencies of both assays without preserving cell-level pairing. Copying a source joint distribution then confounds association with cohort-specific abundance.

Results: We predict the missing recipient joint table by estimating its centered log-linear interaction at fixed margins. Penalized exact conditional likelihood estimates the population interaction and shrinks donor deviations; the estimate is then combined with each recipient's observed margins. Fitted on 36 Cambridge donors, the method achieved mean deviance per cell 0.01220 across 56 Newcastle donors, compared with 0.01478 for signed-root deviance-statistic transfer. The reduction was 17.46% (paired-donor 95% confidence interval for the loss difference, -0.00413 to -0.00080), with lower loss in 50/56 donors. Mean deviance for the destroyed-link control was 0.02274, and the intact-link method had lower loss in 51/56 donors. Across 128 ground-truth simulations with heterogeneous source interactions, donor shrinkage reduced deviance by 1.85% (1.52% to 2.18%) relative to common-effect conditional estimation. Under strong margin shift, fixed-margin prediction reduced deviance by 65.49% (63.02% to 67.72%) relative to signed-statistic transfer.

Conclusions: Fixed-margin prediction separates cross-modal association from cohort-specific abundance. Across held donors and ground-truth simulations, recipient margins corrected abundance shift, whereas donor shrinkage helped when source interactions were heterogeneous.

Keywords: single-cell multi-omics; conditional likelihood; log-linear association; transfer learning; CITE-seq; reproducibility

1 Background

Linked single-cell assays measure RNA, chromatin, proteins, molecular age, and lineage in the same cell or experimentally related unit [1–6]. Joint latent-variable models integrate paired measurements or predict a missing modality. We ask whether cross-modal association learned from linked source donors can predict a recipient joint distribution from its two observed margins.

Marginal frequencies change with cell composition, activation, sequencing depth, and assay scale. Copying a source table therefore transfers abundance together with dependence. Signed Pearson and likelihood-ratio statistics summarize departure from independence, but their scale and attainable range depend on the margins. A statistic pooled across donors can consequently change meaning at recipient margins.

Log-linear interactions isolate association in contingency tables [7–10]. Conditional likelihood removes row and column nuisance parameters, and common-effect or mixed-effects models share odds-ratio information across observed strata [11–13]. Copula and differential-association methods also estimate dependence from observed single-cell or multi-omic pairings [14–17]. These methods estimate dependence where pairings are observed; our target is the unobserved joint table at new margins.

Centered log-linear interactions and conditional odds-ratio estimation are established tools for observed contingency tables. We use them to predict an unobserved recipient table from linked source donors and the recipient margins. The method pools heterogeneous donor effects, then reconstructs each table at the abundance observed in its recipient. We prove that the penalized fit is unique and bound recipient risk by source estimation error and recipient drift. Ground-truth simulations separate margin shift from donor heterogeneity. Public CITE-seq studies compare the method with signed-statistic transfer, fitted interaction models, and margin-preserving link destruction.

2 Methods

2.1 Association coordinates

For ordered marker pair e, let P_e be a positive joint-probability table for linked states U and V. Orthonormal contrast matrices B_U and B_V remove the row and column components of its log table. We call the resulting centered log-linear interaction the *coupling field*:

$$\Theta(P_e) = B_U^\mathsf{T} \log(P_e) B_V \tag{1}$$

where the logarithm is elementwise. This field contains all log-linear interaction degrees of freedom, is invariant to positive row or column tilts, and vanishes exactly at independence.

For a binary table, Eq. (1) is one-half of the log odds ratio,

$$\Theta(P_e) = \frac{1}{2} \log \frac{P_{e,00} P_{e,11}}{P_{e,01} P_{e,10}} = \frac{\theta_e}{2}. \tag{2}$$

Proposition S1 in Additional file 1 proves these properties.

2.2 Hierarchical conditional estimator

Let A_{de} be the upper-left count in the binary table for donor d and marker pair e, and let m_{de} denote its margins. Conditional on these margins, the table has one free count. Its exact distribution is

$$p(a \mid m_{de}, \theta) = \frac{w_{de}(a) \exp(\theta a)}{\sum_{b \in \mathcal{A}_{de}} w_{de}(b) \exp(\theta b)}, \tag{3}$$

where $w_{de}(a)$ is the central-hypergeometric combinatorial weight. The distribution depends on the log odds θ but not on row or column effects.

We write the donor-specific log odds as a population effect μ_e plus a donor deviation δ_{de}. The estimator minimizes

$$\begin{aligned} \mathcal{L}(\mu, \delta) = & -\frac{1}{D} \sum_{d,e} \log p(A_{de} \mid m_{de}, \mu_e + \delta_{de}) \\ & + \frac{\gamma}{2} \sum_{d,e} \delta_{de}^2 + \frac{\rho}{2} \|\mu\|_2^2 + \frac{\tau}{2} \mu^\mathsf{T} L \mu. \end{aligned} \tag{4}$$

The first term is the donor-normalized negative exact conditional log likelihood summed over marker pairs. The panel objective is composite because marker pairs from one donor reuse cells. The parameter γ shrinks donor deviations toward the population interaction, ρ penalizes the population interaction, and τ controls optional smoothing by the source-derived positive-semidefinite graph Laplacian L. Penalty scaling, optimization, and convergence criteria are given in Additional file 1.

Conditioning a Poisson log-linear model on both margins removes every row and column parameter and yields Eq. (3). Positive donor and population penalties give Eq. (4) a unique finite minimizer; Proposition S2 in Additional file 1 gives the derivation and proof.

For recipient q, the prediction is the fixed-margin expected table at transported population log odds,

$$\widehat{N}_{qe} = \mathbb{E}_{\alpha \widehat{\mu}_e}[N_{qe} \mid m_{qe}], \tag{5}$$

where the development design selects α from a prespecified study grid. The binary cohort studies used $\{0.5, 0.75, 1, 1.25\}$, and the longitudinal calibration also included zero. This table preserves the recipient margins without a pseudocount or post-fit projection.

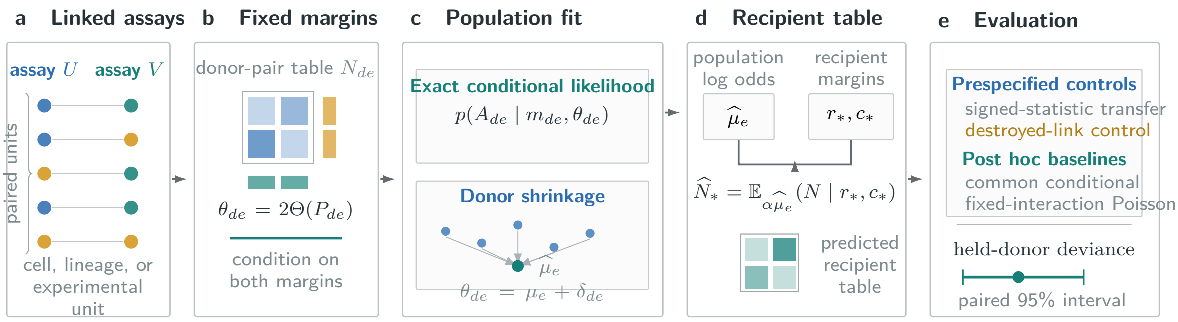

Fig. 1 Conditional transport of pairwise association. Linked measurements define donor-level joint tables. Conditioning on both margins isolates log-linear association, and donor shrinkage estimates the population component shared across donors. Combining this estimate with recipient margins reconstructs a feasible recipient table. Held comparisons use signed-statistic transfer and margin-preserving link destruction. Common-effect conditional and fixed-interaction Poisson baselines were evaluated post hoc.

Proposition S3 bounds expected excess conditional log loss by fixed-margin Fisher information times the sum of squared source transport error and recipient drift variance. Source data can reduce the first term, whereas recipient drift determines the residual floor. Additional file 1 gives the proof and a finite-support upper bound. We assessed held reconstructed tables separately by multinomial deviance.

For the longitudinal calibration, the population log odds varied with visit time, and each patient at each time point had a penalized deviation. A chain-Laplacian penalty coupled adjacent time points; an optional marker-pair hypergraph penalty played the role of L in Eq. (4). Leave-one-patient-out calibration selected the penalties and transport multiplier from losses averaged within each omitted patient and then across patients. The direct comparator fitted time-specific interaction coefficients by ridge-penalized profiled Poisson likelihood under the same folds and transport grid. Additional file 1 gives the complete models and selection grids.

2.3 Classical comparisons and controls

The matched statistic comparator transfers either the signed root Pearson statistic or the signed root likelihood-ratio deviance from the row-plus-column independence model. We normalized statistics by table size, pooled them across source donors, rescaled them for each recipient, and inverted them at its margins. Each study fixed its pooling rule in advance; development data selected the statistic, any evaluated null centering, and the transport multiplier. Both methods received the same recipient margins without recipient pairings.

Two fitted interaction baselines compared alternative pooling rules. The common-effect conditional maximum-likelihood estimate fitted one unpenalized log odds per marker pair across donor strata. The pooled-table baseline used the sample log odds ratio of the donor-pooled table. In the post hoc comparison, both coefficients were reconstructed by fixed-margin conditional expectation with a development-selected transport multiplier. We call the second method pooled-table log odds with conditional reconstruction.

A separate classical baseline treated the pooled-table coefficient as a fixed interaction in a saturated Poisson log-linear model. After development-only selection of the transport multiplier, we refitted recipient row and column parameters at that interaction. The resulting continuous table has the recipient margins and transported odds ratio. This construction differs from the conditional noncentral-hypergeometric expectation used above.

The destroyed-link control permuted complete protein profiles within each donor before joint tables were formed. This preserved RNA and protein margins and dependence within the protein panel but removed cell-level RNA-protein linkage. We refitted the selected conditional estimator to the permuted tables. Fixed-margin independence and a graph-zero fit provided further controls.

For observed table T, positive prediction $\widehat{T}$, and table total n, the loss is multinomial deviance per cell,

$$D(T, \widehat{T}) = \frac{2}{n} \sum_{i,j:T_{ij}>0} T_{ij} \log \frac{T_{ij}}{\widehat{T}_{ij}}. \tag{6}$$

Entity losses were averaged within each evaluation unit. Confirmatory means weighted physical donors equally. The 95% intervals used 20,000 paired bootstrap draws at the donor or prespecified batch-block level, conditional on the source fit, selected configuration, and deterministic cell sample. Development intervals resampled the displayed samples or batches.

2.4 Ground-truth simulation

We simulated 128 replicates from the exact fixed-margin model with known population log odds, 12 source donors, 12 recipient donors, 16 marker pairs, and 128 observations per table. Source donor effects were either homogeneous or normally heterogeneous with standard deviation 1.2. Recipient margins either followed the source distribution or underwent a strong shift. Hyperparameters and transport multipliers were fixed across all four scenarios. We compared the hierarchical estimator with the common-effect conditional estimate and raw and null-centered signed-root deviance-statistic transfer. Paired bootstrap intervals resampled simulation replicates in 20,000 draws.

2.5 Prespecified held evaluation

The Stephenson held-site confirmation followed an executable public plan released before the linked evaluation outcomes were accessed. It fixed eligibility, donor allocation, cell selection, state definitions, marker panels, estimator grids, comparators, loss, uncertainty, and success criteria. After the pilot passed, all development donors supplied the final fit. Predictions from the recipient margins and their checksums were published before the recipient RNA-protein pairings were accessed.

Confirmation required at least 5% lower mean deviance than both the selected statistic-transfer and destroyed-link controls, a paired-bootstrap upper endpoint below zero, and a one-sided sign-test $P \leq 0.025$. Studies that failed data-support or pilot criteria did not proceed to held evaluation. No configuration was revised after release of its plan.

2.6 Datasets and state construction

Stephenson et al. profiled PBMC RNA and surface proteins with a common TotalSeq-C panel across Cambridge and Newcastle [18]. The split used 12 Cambridge calibration donors, 24 Cambridge pilot donors, and 56 Newcastle donors for the held-site test. Babdor et al. profiled healthy baseline participants in seven acquisition pools with the same chemistry and antibody panel [19]. Twelve DB1 donors formed the calibration set, and 20 donors from DB1 and DB2 formed the pilot. The analysis did not proceed beyond the pilot because its criterion was not met. Both studies used a primary panel of 81 ordered pairs from nine cognate RNA and protein markers. The Babdor analysis also fixed a secondary panel of 576 pairs from 24 markers.

Each donor contributed 512 cells selected by a deterministic rule that did not access RNA or ADT counts. RNA state recorded raw detection. Deterministic within-donor protein ranks assigned 256 low and 256 high cells, fixing every protein margin before evaluation.

The GSE239452 pregnancy study used seven nonpregnant calibration donors, eight nonpregnant pilot donors, and nine pregnant evaluation donors [20]. We report it as a post-access corrected cohort replication.

GSE317605 supplied an independent longitudinal PBMC CITE-seq calibration panel from a phase II biliary-cancer trial, with seven patients measured at four visits and 16 matched RNA and ADT markers [21]. Leave-one-patient-out calibration compared the field estimator directly with tuned, time-conditioned, ridge-penalized Poisson prediction. Because the criterion was not met, pilot and held count matrices were not accessed.

COMBAT CITE-seq supplied 12 Oxford calibration and 24 Oxford pilot samples [22]. The BMMC development analysis used two fit donors and four batches from a third donor. These studies provided adaptive comparisons of the conditional estimator with signed-statistic transfer. Table 1 gives the role of each analysis.

2.7 Software and reproducibility

The tagged public release contains the estimator and cohort reducers, the analysis plans and held predictions, the derived cohort results, the programs for the ground-truth simulation and risk

Table 1 Binary conditional-transfer datasets and evaluation roles.

Study	Transfer	Fit units	Evaluation units	RNA-ADT pairs	Evidence role
COMBAT	Oxford calibration to Oxford pilot	12	24	81	development
BMMC	two fit donors to bridge donor batches	2 donors	4 batches	100	development
Stephenson	Cambridge to Newcastle	36 donors	56 donors	81	pre-outcome held site
GSE314416	DB1 calibration to DB1 and DB2 pilot	12 donors	20 donors	81; 576 secondary	pilot criterion failed
GSE239452	nonpregnant to pregnant	15 donors	9 donors	81	post-access corrected replication
GSE317605	leave-one-patient-out longitudinal calibration	7 patients	7 patients (28 visits)	256	calibration criterion failed

stress test, and verification tests. OpenAI Codex assisted with code development, literature retrieval, manuscript editing, and LaTeX preparation.

3 Results

3.1 Conditioning and donor shrinkage separated distinct errors

The factorial simulation separated donor heterogeneity from margin shift (Fig. 2). With heterogeneous donor effects and source-like margins, the hierarchical estimator reached deviance 0.01493, compared with 0.01521 for the common-effect conditional estimate. The reduction was 1.85% (95% interval 1.52% to 2.18%). After the recipient margins shifted, conditional transport reduced deviance by 65.49% (63.02% to 67.72%) relative to raw signed-root deviance-statistic transfer.

Under homogeneous donor effects and source-like margins, common-effect conditional deviance was 0.008660, compared with 0.008752 for the hierarchical estimator, a shrinkage cost of 1.06%. After margin shift, the two conditional estimators remained close (0.008640 and 0.008674), whereas statistic-transfer deviance rose to 0.02943. Fixed-margin reconstruction therefore absorbed the marginal shift in this model; donor shrinkage contributed only when source associations varied across donors.

The finite-support calculations satisfied both risk bounds throughout the prespecified stress grid. Under unbiased transport, risk increased monotonically with recipient drift in every margin family (Additional file 1).

3.2 Association transferred across recruitment sites

Across 56 Newcastle donors, the hierarchical conditional estimator reached mean deviance 0.01220, compared with 0.01478 for signed-root deviance-statistic transfer. The corresponding 17.46% reduction had a paired-donor loss-difference interval of -0.00413 to -0.00080. The conditional estimate was better in 50/56 donors ($P = 5.09 \times 10^{-10}$, one-sided sign test). All 36 Cambridge development donors supplied the final source fit after pilot selection.

The estimator also reduced deviance by 46.36% relative to the destroyed-link fit of 0.02274 (paired-donor interval -0.01323 to -0.00749; 51/56 donors favorable). In post hoc fitted-baseline analyses, it improved on the common-effect conditional estimate by 6.18% (relative-reduction interval 3.84% to 8.47%) and on pooled-table log odds with conditional reconstruction by 3.39% (1.42% to 5.37%).

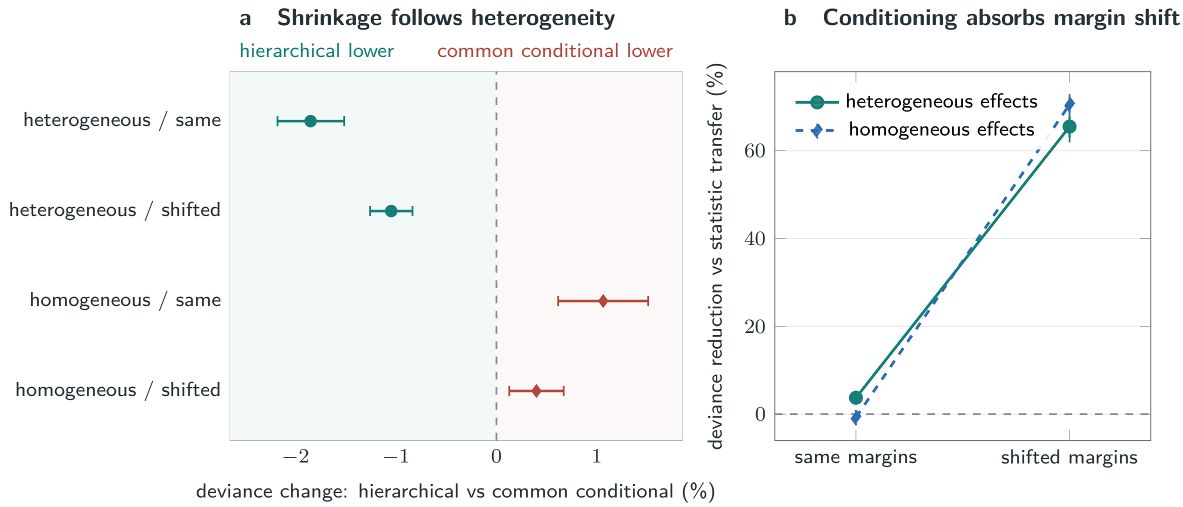

Fig. 2 **Ground-truth estimator behavior.** **a**, relative change in recipient deviance for the hierarchical estimator against the common-effect conditional estimate. Negative values favor hierarchical shrinkage. **b**, reduction in deviance relative to raw signed-root deviance-statistic transfer before and after recipient-margin shift. Bars are 95% paired simulation-replicate bootstrap intervals across 128 replicates.

3.3 The corrected cohort favored conditional reconstruction

GSE239452 was analyzed as a post-access corrected cohort replication. The hierarchical conditional estimate reached mean deviance 0.00851, compared with 0.01413 for corrected signed-root deviance-statistic transfer, a 39.81% reduction favored by all nine pregnant donors. The common-effect conditional estimate was 2.20% lower than the hierarchy and was favored in all nine donors. Additional file 1 reports the fitted-interaction and fixed-interaction Poisson comparisons.

3.4 Longitudinal calibration stopped before held evaluation

Across 7 GSE317605 patients and 28 visits, leave-one-patient-out calibration selected a zero hypergraph penalty. Mean deviance was 0.006530, compared with 0.006609 for tuned, time-conditioned, ridge-penalized Poisson prediction. The 1.19% reduction favored the field estimate in 6/7 patients and 4/4 time points. Progression required at least 5% improvement over Poisson and strict improvement over a separately retuned graph-zero fit. Because the selected primary and graph-zero configurations were identical, neither condition passed and no pilot or held matrix was requested. Relative to destroyed links, the reduction was 17.39%, with favorable results in 7/7 patients.

4 Discussion

Recipient margins and source heterogeneity govern different parts of the prediction. Fixed-margin reconstruction absorbed abundance shifts in simulation and transferred RNA-protein association from Cambridge to Newcastle without copying source composition. Donor shrinkage improved on common-effect conditional estimation at the held site and under simulated heterogeneity. Common-effect pooling was stronger in the corrected pregnancy cohort and under homogeneous simulated effects. The value of hierarchical pooling therefore depends on variation in source association.

Classical stratified odds-ratio models estimate association among observed tables. Our target is an unobserved recipient table constrained by its observed margins. The method combines hierarchical donor pooling with fixed-margin reconstruction, and the held-site design tests the resulting prediction under criteria set before outcome access. Comparisons with common-effect, pooled-table, and fixed-interaction Poisson baselines show that accuracy depends on the interaction coordinate and on how it is pooled across donors.

Table 2 Hierarchical conditional transfer versus the selected signed-statistic method.

Study	Role	Conditional loss	Statistic-transfer loss	Reduction	Difference 95% CI	Favorable
COMBAT	development	0.01156	0.01561	25.94%	-0.00597 to -0.00225	17/24
BMMC	development	0.01085	0.01315	17.47%	-0.00504 to -0.00038	4/4
Stephenson	held site	0.01220	0.01478	17.46%	-0.00413 to -0.00080	50/56
GSE239452	post-access corrected replication	0.00851	0.01413	39.81%	-0.00710 to -0.00423	9/9

Loss is the evaluation-unit-equal mean multinomial deviance per cell; differences are conditional minus statistic transfer. Held-study units are physical donors. Development units are samples for COMBAT and batches for BMMC; the four BMMC batches come from one bridge donor.

5 Limitations

The held confirmation and corrected cohort replication are observational RNA-protein studies and do not establish perturbation causality or a molecular mechanism. Both use binary states, 512 cells per donor, and the same nine RNA and nine protein markers. The 81 pairwise tables within a donor reuse cells and marker vectors. The estimand is conditional dependence, not zero-shot protein abundance. Broader panels, technologies, tissues, state resolutions, and marker sets require independent replication.

The selected graph penalty was zero in COMBAT, BMMC, Stephenson, GSE239452, and the GSE317605 calibration, so the held evidence covers graph-zero hierarchical conditional transfer. It does not establish graph or hypergraph regularization. The head-to-head comparisons combine conditional estimation, donor shrinkage, pooling, reconstruction, and boundary handling; the held studies do not identify each contribution separately. The fixed-interaction Poisson analysis was post hoc. A post hoc Stephenson sensitivity retained 29 of 32 repeated destroyed-link fits because three failed the numerical criterion. COMBAT and BMMC are adaptive development analyses, and BMMC's four evaluation units are batches from one bridge donor. Their intervals are descriptive.

The Stephenson sign test is study specific and does not correct familywise error across the preceding public-study search. Public plans prevented outcome-guided reruns within each study, but the sequence selected datasets that passed file, feature, support, and development gates. GSE179221 stopped on the first source donor because its prespecified cognate feature axis was not unique; no model or held outcome was accessed.

An exploratory multistate extension depends on discretization, a pseudocount, and finite permutation centering. None of its seven public panels met the complete replication and strongest-comparator criteria; full results are reported in Additional file 1. Spatial applications require linked observations because unpaired regional summaries do not supply the joint tables required by the estimand.

6 Conclusions

Fixed-margin interaction prediction reconstructs cross-modal association at recipient-specific abundance. It outperformed prespecified signed-statistic and destroyed-link controls in a held-site experiment. Recipient margins address abundance shift; source heterogeneity determines the value of donor shrinkage.

Abbreviations

ADT: antibody-derived tag; PBMC: peripheral blood mononuclear cell; RNA: ribonucleic acid.

Declarations

Ethics approval and consent to participate

This study used only public, de-identified data and involved no new participants, specimens, interventions, or identifiable records; the original studies report their ethics approvals and consent procedures.

Consent for publication

Not applicable.

Availability of data and materials

All source datasets are public through the accessions listed in Additional file 1 and retain their original licenses. Analysis plans, MIT-licensed software, cohort and simulation results, study records, and verification scripts are available in the immutable v2.0.3 release at https://github.com/sushaan-k/coupling-fields-benchmark/releases/tag/coupling-fields-v2.0.3-public-benchmark. Raw matrices are not redistributed. The platform-independent software requires Python 3.9 or newer and the version-pinned NumPy and SciPy dependencies in the release; there are no additional restrictions on use.

Competing interests

Competing-interest information is withheld in the blinded manuscript.

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Authors' contributions

Author names and contribution statements are withheld in the blinded manuscript.

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Additional file

Additional file 1 (PDF). *Mathematical results and public benchmark details.* Proofs, algorithms, dataset provenance, study status, comparator definitions, sensitivity analyses, and reproducibility information.

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